How Brains Learn KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: conditioning food sound stronger weight wire

You can use a word more than once.

1. Your other hand. Write your name with your other hand. Your partner times you.

	Try 1	Try 2	Try 3
Time (seconds)	Answers vary.		

2. Stronger connections.

When you practice, the synapses between your neurons get stronger.

"Neurons that fire together, wire together."

3. Pavlov's dogs.

Pavlov's team played a sound (a metronome) right before they gave the dogs food.

Soon the dogs drooled when they heard the sound, even with no food.

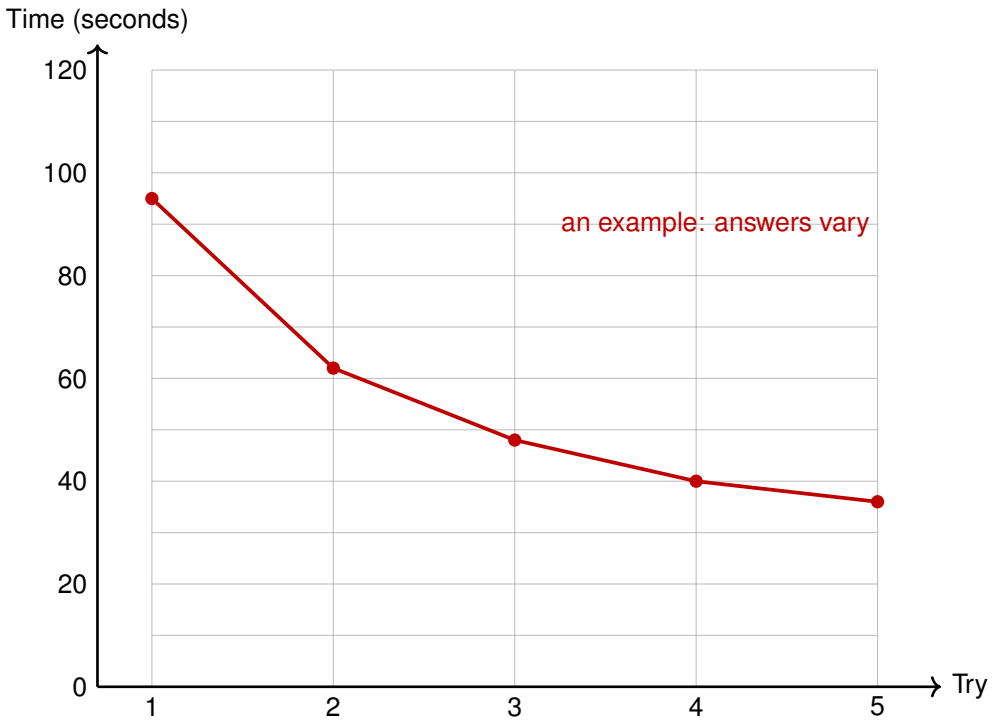
Learning to connect two things that happen together is called conditioning.

In AI, the strength of a connection is a number called a weight.

4. Mirror drawing. Use the same pen for all 5 tries.

	Try 1	Try 2	Try 3	Try 4	Try 5
Time (seconds)	Answers vary.				
Mistakes					

5. My learning curve. Put a dot for each try. Connect the dots.



What happened to your time? It dropped fast at first, then more slowly.

6. Talk about it. Your brain changes synapses. AI changes weights. Is it the same kind of learning?

Brain learning and AI learning are **the same** / **different** because Answers vary.

A sound that makes me reach for my phone: Answers vary. That is like Pavlov's dogs.

Go Further (Day 2). A new mirror: left and right flipped.

Same pen as yesterday: _____ seconds. New mirror: _____ _____ _____ seconds.

Was it hard again? yes Why? My brain learned one skill, not every mirror skill.